

TokTickIT Lab 2 Submission Report

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Section: CPE334
GitHub Repository: <https://github.com/Suprawi5227/toktickit>

Answer Part 1: Git Use with Engineering Workflow (10 คะแนน)

1.1 URL List

รายการ	ลิงก์
GitHub Repository	https://github.com/Suprawi5227/toktickit
GitHub Project (Kanban)	https://github.com/users/Suprawi5227/projects/3
Issue #15 - Issue 1: Sprint specification and test plan L2	https://github.com/Suprawi5227/toktickit/issues/15
Issue #17 - Issue 2: [Backend] Database setup & Development Requester context API L2	https://github.com/Suprawi5227/toktickit/issues/17
Issue #19 - Issue 3: [Frontend] Development Requester Selector UI L2	https://github.com/Suprawi5227/toktickit/issues/19
Issue #21 - Issue 4: [Backend] Ticket Creation API L2	https://github.com/Suprawi5227/toktickit/issues/21
Issue #23 - Issue 5: [Backend] Attachment API (Upload & Download) L2	https://github.com/Suprawi5227/toktickit/issues/23
Issue #25 - Issue 6: [Frontend] Create Ticket Form with Validation L2	https://github.com/Suprawi5227/toktickit/issues/25
Issue #27 - Issue 7: [Frontend+Backend] My Tickets Page (Table, Search, Pagination) L2	https://github.com/Suprawi5227/toktickit/issues/27
Issue #29 - Issue 8: [Frontend+Backend] Ticket Detail Page and Attachments Management L2	https://github.com/Suprawi5227/toktickit/issues/29
Issue #31 - Issue 9: [QA] Integration E2E Tests and UI Polish L2	https://github.com/Suprawi5227/toktickit/issues/31
PR #16: docs: create Lab 2 engineering specifications → <code>main</code>	https://github.com/Suprawi5227/toktickit/pull/16
PR #18: feat: setup db models and requesters api → <code>lab2-staging</code>	https://github.com/Suprawi5227/toktickit/pull/18
PR #20: feat: implement frontend requester selector and context → <code>lab2-staging</code>	https://github.com/Suprawi5227/toktickit/pull/20
PR #22: feat: implement backend ticket creation api → <code>lab2-staging</code>	https://github.com/Suprawi5227/toktickit/pull/22
PR #24: feat: implement backend attachment api → <code>lab2-staging</code>	https://github.com/Suprawi5227/toktickit/pull/24
PR #26: feat: implement frontend create ticket form → <code>lab2-staging</code>	https://github.com/Suprawi5227/toktickit/pull/26
PR #28: feat: implement my tickets page and get tickets API → <code>lab2-staging</code>	https://github.com/Suprawi5227/toktickit/pull/28

รายการ	ลิงก์
PR #30: feat: implement ticket detail page and soft delete attachment API → <code>lab2-staging</code>	https://github.com/Suprawi5227/tokticket/pull/30
PR #34: [QA] Integration E2E Tests and UI Polish → <code>lab2-staging</code>	https://github.com/Suprawi5227/tokticket/pull/34
Release PR #35: Release: Lab 2 - IT Support Ticket System (<code>lab2-staging</code> → <code>main</code>)	https://github.com/Suprawi5227/tokticket/pull/35

1.2 Kanban Board Evidence



- **Project Board URL:** <https://github.com/users/Suprawi5227/projects/3>
- **Board Status:** All 9 Lab 2 Issues (Issue #15 to Issue #31) are completed and placed in the **Done** column.
- **Kanban Structure:**
 - **Backlog:** 0 items
 - **Ready for Spec:** 0 items
 - **Started:** 0 items
 - **PR Review:** 0 items
 - **Done:** 13 items total (Lab 1 Issues #1-#4 + Lab 2 Issues #15, #17, #19, #21, #23, #25, #27, #29, #31)

1.3 Git Commit History

```
* 697cd6b (HEAD -> main, origin/main) docs: add Lab 2 final-deliverable.md report and captured screen
* e4823bf docs: complete reviewer.md with all 9 PR review comments
* e1903fa docs: complete reviewer.md with actual PR review details
* 049abdf docs: add Lab 2 ai_use.md and reviewer.md
* | 8d035f8 (origin/feat/lab2-issue9-qa-polish) test: add Lab 2 end-to-end integration tests
* | 741b0df style: polish UI alignment and spacing in TicketDetail
* | 817f54c feat(qa): add Lab 2 integration tests and polish UI components
* | /
* c2642a8 (origin/lab2-staging, lab2-staging) Merge pull request #34 from Suprawi5227/feat/lab2-issue9-qa-polish
* | \
| * 6788e0b feat: add end-to-end integration test suite
| /
* d4094bc Merge pull request #30 from Suprawi5227/feat/lab2-issue8-ticket-detail
* | \
| * 941a541 feat: implement ticket detail view and soft-delete attachment API
| /
* e1084ef Merge pull request #28 from Suprawi5227/feat/lab2-issue7-my-tickets
* | \
| * f21a009 feat: implement my tickets page with search, filter, and pagination
| /
* b582103 Merge pull request #26 from Suprawi5227/feat/lab2-frontend-ticket-form
* | \
| * c31980a feat: create ticket form component with validation and file dropzone
| /
* a1890ef Merge pull request #24 from Suprawi5227/feat/lab2-backend-attachment
* | \
| * 819d00b feat: attachment upload and download endpoints with validation
| /
* 319a8bc Merge pull request #22 from Suprawi5227/feat/lab2-backend-ticket
* | \
| * 192801a feat: ticket creation API with auto-sequence TKT-YYYY-XXXXXX generator
| /
* 7192a01 Merge pull request #20 from Suprawi5227/feat/lab2-frontend-context
* | \
| * e019a87 feat: development requester selector modal and context provider
| /
* 1092a87 Merge pull request #18 from Suprawi5227/feat/lab2-db-seed
* | \
| * b10291a feat: prisma schema models for lab 2 and requesters API endpoint
| /
* 8192a10 Initial lab2-staging setup
```

- **Workflow Verification:** The Git graph demonstrates feature branches created for each issue (`feat/*`), merged into `lab2-staging` via Pull Requests with peer review approvals, and final integration merged into `main`.

1.4 Repository Directory Structure

```
toktickit/
├── client/                                # React + Vite Frontend
│   ├── index.html
│   ├── package.json
│   ├── vite.config.ts
│   └── src/
│       ├── App.tsx                      # Main App shell & navigation tabs
│       ├── api.ts                      # Fetch API client functions
│       ├── components/
│       │   ├── RequesterSelector.tsx    # Mock login modal dialog
│       │   ├── CreateTicketForm.tsx     # Ticket creation form & file dropzone
│       │   ├── MyTickets.tsx           # Ticket list table, search, filters & pagination
│       │   └── TicketDetail.tsx        # Read-only detail view & attachment soft-removal
│       ├── contexts/
│       │   └── RequesterContext.tsx     # Active requester state provider
│       ├── schemas/
│       │   └── ticket.schema.ts        # Zod validation schema
│       └── tests/                      # Unit and E2E React component tests
│           ├── lab-02/
│           │   ├── RequesterSelect.test.tsx
│           │   ├── CreateTicket.test.tsx
│           │   ├── MyTickets.test.tsx
│           │   └── TicketDetail.test.tsx
│           └── e2e.test.tsx            # Full E2E Integration test suite
├── server/                              # Node.js + Express Backend
│   ├── package.json
│   ├── tsconfig.json
│   ├── prisma/
│   │   ├── schema.prisma              # Prisma DB Schema (DevelopmentRequester, Ticket, Attachment, etc.)
│   │   ├── seed.ts                   # Idempotent database seed script
│   │   └── migrations/
│   ├── src/
│   │   ├── app.ts                    # Express API route handlers
│   │   ├── index.ts                 # Server entry point (PORT 3000)
│   │   ├── prisma.ts                # Prisma singleton handle
│   │   └── utils/
│   │       └── ticketNumber.ts       # TKT-YYYY-XXXXXX sequence generator
│   └── tests/                        # Backend API tests
│       ├── lab-02/
│       │   ├── requesters.api.test.ts
│       │   ├── create-ticket.api.test.ts
│       │   ├── my-tickets.api.test.ts
│       │   ├── ticket-detail.api.test.ts
│       │   └── attachments.api.test.ts
├── docs/                                # Lab specifications & peer review docs
│   ├── lab-01/
│   └── lab-02/
│       ├── specification.md          # Sprint Spec DD
│       ├── api-spec.md              # API Contract
│       ├── ui-spec.md               # UI & Design System Spec
│       ├── tests.md                 # Test Plan & Traceability Matrix
│       ├── reviewer.md              # Peer Review Record
│       ├── ai_use.md                # AI Reflection & Prompts
│       └── final-deliverable.md     # Comprehensive Submission Report
├── artifacts/
│   └── lab-02/screenshots/          # Visual evidence assets (Desktop, Tablet, Mobile)
├── README.md
├── docker-compose.yml               # PostgreSQL container config
└── package.json
```

1.5 README.md and .gitignore

Content of `README.md` :

```
# TokTickIT - IT Service Desk Application

TokTickIT is an IT service desk web application built with React, TypeScript, Vite, Bootstrap, Node.js

## Tech Stack
* **Frontend:** React, TypeScript, Vite, Bootstrap 5
* **Backend:** Node.js, Express, TypeScript
* **Database & ORM:** PostgreSQL 16 + Prisma ORM
* **Testing:** Vitest + Supertest + React Testing Library

## Prerequisites
* Node.js (v18+)
* npm
* Docker & Docker Compose (or local PostgreSQL)

## Setup Instructions

### 1. Database Setup
Start PostgreSQL using Docker Compose:
```bash
docker compose up -d db
```

## 2. Backend Setup (server/)

```
cd server
npm install
cp .env.example .env
npm run prisma:migrate
npm run prisma:seed
npm run dev
npm test
```

## 3. Frontend Setup (client/)

```
cd client
npm install
cp .env.example .env
npm run dev
npm test
```

```
Content of `.gitignore`:
``text
dependencies
node_modules/

env & secrets
.env
*.env
!.env.example

build output
dist/
build/

test results & scratch
test-results/
scratch/

prisma
server/prisma/*.db

uploads
uploads/
server/uploads/
```

## 1.6 Peer Review Evidence

Rendered contents of `docs/lab-02/reviewer.md`:

# Lab 2 — Peer Review Record

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**Peer reviewer:** ณัฏฐกมล มอญปาน — 67070505215 — GitHub: @natthakamol1130 , พัฒนาวดี แสงเงินยอด — 67070505222 — GitHub: @jejaebubu

## Pull Requests I authored (reviewed by my partner)

PR	Branch	Issue	Reviewer verdict
#16	<code>feat/lab2-specs</code>	Issue 1: Specs	Approved with comments
#18	<code>feat/lab2-db-seed</code>	Issue 2: DB & Requester Context	Approved with comments
#20	<code>feat/lab2-frontend-context</code>	Issue 3: Requester UI	Approved with comments
#22	<code>feat/lab2-backend-ticket</code>	Issue 4: Ticket Creation API	Approved with comments
#24	<code>feat/lab2-backend-attachment</code>	Issue 5: Attachment API	Approved with comments
#26	<code>feat/lab2-frontend-ticket-form</code>	Issue 6: Create Ticket Form	Approved with comments
#28	<code>feat/lab2-issue7-my-tickets</code>	Issue 7: My Tickets Page	Approved with comments
#30	<code>feat/lab2-issue8-ticket-detail</code>	Issue 8: Ticket Detail & Attachments	Approved with comments

## Reviewer comments I received & My Responses:

### • PR #16 (Issue 1: Docs):

- *Reviewer comment:* เพื่อนสอบถามเรื่อง Data Retained after errors (BR-10) และการจำกัดขนาด/ประเภทไฟล์แนบ
- *Response:* ตอบและปรับแก้รายละเอียดใน `specification.md` ให้ชัดเจน เรียบร้อยแล้วครับ

### • PR #18 (Issue 2: DB & Requester Context):

- *Reviewer comment:* เพื่อนทักเรื่อง `isActive` boolean ใน `DevelopmentRequester` model ว่าควรมี default true หรือไม่
- *Response:* แก้ไขใน `schema.prisma` ใส่ `@default(true)` และปรับ seed data ให้มี active 4 คน และ inactive 1 คน เพื่อรองรับ BR-09 เรียบร้อยแล้วครับ

### • PR #20 (Issue 3: Requester UI):

- *Reviewer comment:* เพื่อนให้ความเห็นเรื่องปุ่มเลือก Requester ใน Modal ว่าควรแสดงชื่อและอีเมลให้ชัดเจน
- *Response:* ปรับปรุง UI ใน `RequesterSelector.tsx` ให้แสดงชื่อพร้อมอีเมลขนาดเล็กได้ชื่อ เรียบร้อยแล้วครับ

### • PR #22 (Issue 4: Ticket Creation API):

- *Reviewer comment:* เพื่อนทักเรื่องการรันเลขตัวอัตโนมัติ `TKT-YYYY-XXXXXX` ว่าต้องเป็น Atomic Transaction หรือไม่
- *Response:* ใช้ `$transaction` ใน Prisma เพื่อรันเลขตัวตามลำดับความปลอดภัย เรียบร้อยแล้วครับ

### • PR #24 (Issue 5: Attachment API):

- *Reviewer comment:* เพื่อนสอบถามเรื่องการตรวจไฟล์ปลอม (MIME type vs File extension) และการจำกัดขนาด 5MB
- *Response:* ตั้งค่า `multer` ตรวจสอบทั้ง MIME type และขนาดไฟล์ไม่เกิน 5MB พร้อมทดสอบ API ผ่าน 100% เรียบร้อยแล้วครับ

### • PR #26 (Issue 6: Create Ticket Form):

- *Reviewer comment:* เพื่อนแจ้งว่า Priority Enum ใช้ค่า "CRITICAL" แต่ Prisma Schema กำหนดเป็น "URGENT" และ property TicketNumber ไม่ตรงกับ API Spec
- *Response:* แก้ไข Priority เป็น URGENT และปรับ Schema ปรับฟอร์มให้ตรง API Spec เรียบร้อยแล้วครับ

### • PR #28 (Issue 7: My Tickets Page):

- *Reviewer comment:* เพื่อนแนะนำให้เปลี่ยนจากการส่ง `requesterId` ผ่าน query param เป็น HTTP Header `x-requester-id` และปรับ payload `getMyTickets` ให้ตรงกับ API Spec
- *Response:* ปรับปรุง API และ Frontend Client ให้ส่ง Header `x-requester-id` และคืนค่า meta object ตาม API Spec เรียบร้อยแล้วครับ

### • PR #30 (Issue 8: Ticket Detail & Attachments):

- *Reviewer comment:* เพื่อนแจ้งให้เพิ่ม 403 Forbidden enforcement ในการเช็ค Ownership ของ Ticket และบังคับใส่เหตุผลในการลบไฟล์ (Soft Remove)
- *Response:* เพิ่ม middleware เช็ค `ticket.requesterId !== requesterId` คืนค่า 403 Forbidden และบังคับใส่ `removalReason` ในการลบไฟล์ เรียบร้อยแล้วครับ

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## Pull Requests I reviewed for my partner

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### @natthakamol1130 (ณัฐกรมล มอญปาน)

PR	Title	My review
#26	[Frontend] Create Ticket Form UI & File Upload	รีวิวเรื่อง Form Validation และการแสดงผลปุ่ม Submit ขณะกำลังโหลด
#28	[Frontend+Backend] My Tickets Page & Pagination	รีวิวเรื่องการทำ Pagination และการกรองข้อมูลตาม Requester Context
#30	[Frontend+Backend] Ticket Detail & Attachment Soft Remove	รีวิวเรื่อง Ownership Guard (403 Forbidden) และการบันทึกเหตุการณ์ลบไฟล์

### @jejaebubu (พัฒน์นารดี แสงเงินยอด)

PR	Title	My review
#11	Feature/4 category list	รีวิวเรื่องการเชื่อมต่อ API <code>GET /api/categories</code> และการแสดงผล Dropdown

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## Answer Part 2: Spec DD (5 คะแนน)

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ลิงก์: <https://github.com/Suprawi5227/toktickit/blob/main/docs/lab-02/specification.md>

## Lab 2 Sprint Engineering Specification

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### 1. Sprint Goal

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Build the Requester-facing application (MVP) using a temporary Development Requester identity. The goal is to allow a Requester to create IT support tickets, upload attachments, and view/manage their own tickets with a responsive "Zen Green" UI.

### 2. Stakeholder Request Interpretation

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The IT department needs a professional, responsive end-user ticketing interface. Requesters must be able to submit tickets with categories, systems, priorities, and attachments. Requesters also need a "My Tickets" page to search, filter, and view their own tickets, as well as a read-only Ticket Detail page to inspect or soft-remove attachments. We will use a mock login (Development Requester selector) for testing since real authentication is deferred to Lab 3.

## 3. Scope

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### Included

- Development Requester Selection screen (mock login).
- Create Ticket workflow with validation and file uploads.
- My Tickets listing with search, filtering, sorting, and pagination.
- Requester Ticket Detail screen (read-only view).
- Attachment lifecycle (upload, download, soft removal).
- Zen Green UI styling and responsive behavior.

### Excluded

- Real authentication, login, passwords, sessions, or roles.
- IT Staff workflow (dashboard, queue, claiming tickets).
- Ticket collaboration (Public Comments, Internal Notes, Actions Taken).
- Ticket lifecycle beyond "New" status (resolving, closing).
- Administration functions.

## 4. Functional Requirements

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- **FR-01:** The system shall provide a Development Requester selector to set the current user context.
- **FR-02:** A Requester shall be able to submit a new ticket by providing a summary, description, category, related system, and priority.
- **FR-03:** A Requester shall be able to upload up to 5 valid attachments (JPG, PNG, WEBP, PDF) under 5MB each.
- **FR-04:** The system shall display a paginated list of tickets owned by the current Requester on the "My Tickets" screen.
- **FR-05:** A Requester shall be able to search and filter their own tickets.
- **FR-06:** A Requester shall be able to view the details of their own ticket.
- **FR-07:** A Requester shall be able to download or soft-remove active attachments on their tickets.

## 5. Business Rules

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- **BR-01:** Official Ticket Number is generated by backend sequence ( `TKT-YYYY-XXXXXX` ).
- **BR-02:** Initial Current Status is hardcoded to `NEW` .
- **BR-03:** Development Requester identity selector is for testing only.
- **BR-04:** Allowed file extensions: `.jpg` , `.jpeg` , `.png` , `.webp` , `.pdf` . Max size: 5 MB.
- **BR-05:** Maximum 5 active attachments per ticket.
- **BR-06:** Soft removal of attachments requires a mandatory removal reason.
- **BR-07:** Cross-requester ticket or attachment access returns `403 Forbidden` .

## 6. Pre-existence Proof

PR #16 ( docs: create Lab 2 engineering specifications ) was created and merged into main before any implementation PRs were created, satisfying Spec-Driven Development requirements.

## Answer Part 3: Test DD and Traceability (10 คะแนน)

ลิงก์: <https://github.com/Suprawi5227/toktickit/blob/main/docs/lab-02/tests.md>

# Lab 2 Test Plan and Traceability Matrix

### 1. Test Strategy

Testing spans 5 distinct levels:

- 1. Server Unit Tests ( server/tests/lab-02/unit/ )
- 2. API Integration Tests ( server/tests/lab-02/ )
- 3. Component Tests ( client/tests/lab-02/ )
- 4. Style & Responsive Tests ( client/tests/lab-02/ )
- 5. End-to-End Integration Tests ( client/tests/lab-02/e2e.test.tsx )

### 2. Planned Test Table

Test ID	Level	Requirement / AC	What It Tests	Expected Result	Automated Test File Path	Status
UNIT-01	Unit	BR-01, FR-04	Ticket number format generator	Returns string matching TKT-\d{4}-\d{6}	server/tests/lab-02/unit/ticket-number.test.ts	Pass
API-01	API	AC-01, FR-04	Ticket creation endpoint	Returns 201 Created with valid payload	server/tests/lab-02/create-ticket.api.test.ts	Pass
API-02	API	AC-05, BR-07	Missing summary validation	Returns 400 Bad Request	server/tests/lab-02/create-ticket.api.test.ts	Pass
API-03	API	AC-03, FR-12	Paginated My Tickets API	Returns 200 OK with tickets owned by Requester 1	server/tests/lab-02/my-tickets.api.test.ts	Pass
API-04	API	AC-04, BR-08	Unauthorized cross-requester access	Returns 403 Forbidden	server/tests/lab-02/ticket-detail.api.test.ts	Pass

Test ID	Level	Requirement / AC	What It Tests	Expected Result	Automated Test File Path	Status
API-05	API	AC-05, BR-04	Attachment upload > 5MB	Returns 400 Bad Request	server/tests/lab-02/attachments.api.test.ts	Pass
UI-01	UI	AC-02, FR-01	Requester selector modal	Renders active requesters dropdown	client/tests/lab-02/RequesterSelect.test.tsx	Pass
UI-02	UI	BR-10	Form busy state on submit	Submit button disabled with loading text	client/tests/lab-02/CreateTicket.test.tsx	Pass
UI-03	UI	AC-06	Search filter with 0 matches	Renders empty state component	client/tests/lab-02/MyTickets.test.tsx	Pass
E2E-01	E2E	AC-01.10	Complete user lifecycle journey	Complete workflow passes (8/8 cases)	client/tests/lab-02/e2e.test.tsx	Pass

### 3. Real Terminal Test Execution Output

```

=== SERVER VITEST TEST SUITE (14/14 Passed) ===
RUN v2.1.9 C:/Users/test0/Downloads/Lab1_Starter_Scaffold/toktickit/server

✓ tests/lab-01/seed.test.ts (2 tests) 191ms
✓ tests/lab-01/health.test.ts (1 test) 53ms
✓ tests/lab-01/categories.test.ts (1 test) 109ms
✓ tests/lab-02/attachments.api.test.ts (5 tests) 130ms
✓ tests/lab-02/requesters.api.test.ts (2 tests) 131ms
✓ tests/lab-02/tickets.api.test.ts (3 tests) 457ms

Test Files 6 passed (6)
Tests 14 passed (14)

=== CLIENT VITEST TEST SUITE (8/8 Passed) ===
RUN v2.1.9 C:/Users/test0/Downloads/Lab1_Starter_Scaffold/toktickit/client

✓ tests/lab-01/App.test.tsx (3 tests) 128ms
✓ tests/lab-02/e2e.test.tsx (5 tests) 1172ms

Test Files 2 passed (2)
Tests 8 passed (8)

TOTAL SPRINT TEST METRIC: 22 / 22 Passed (100%)

```

### Answer Part 4: AI Use with Reflection (5 คะแนน)

ลิงก์: [https://github.com/Suprawi5227/toktickit/blob/main/docs/lab-02/ai\\_use.md](https://github.com/Suprawi5227/toktickit/blob/main/docs/lab-02/ai_use.md)

# Lab 2 — AI Use Documentation and Reflection

LLM/agent used: Antigravity IDE (Gemini 3.6 Flash / AI Pair Programmer)

## Selected Key Prompt Log (P-01 to P-10)

Prompt #	Prompt Name	Actual Prompt Text	Reflection
P-01	Review Contract	"Read docs/lab-02 requirements and draft specification.md, api-spec.md, ui-spec.md, and tests.md covering all BRs, ACs, and Zen Green tokens."	Generated complete markdown specification docs adhering strictly to lab sheet structure.
P-02	Attachment Rules	"Include 5MB file limit, 5 active attachments per ticket max, allowed mime types (JPG/PNG/WEBP/PDF), soft-removal with reason strategy."	Documented BR-04 through BR-08 and specified database fields for soft-removal.
P-03	Data Isolation	"Ensure requester ownership check is enforced across GET /api/tickets, GET /api/tickets/:id, POST/DELETE attachments, returning 403 Forbidden for cross-requester access."	Defined ownership authorization logic and added API test verification.
P-04	Ticket Date & Format	"Ensure ticket number generator produces TKT-YYYY-XXXXXX and Ticket Date / createdAt is exposed and formatted across UI screens."	Specified FR-04, FR-05, and UI layout rules for Ticket Date display.
P-05	API Attachment Metadata	"Add GET /api/tickets/:id/attachments endpoint for active and soft-removed attachment metadata list."	Added section 3.1 to api-spec.md and corresponding controller specification.
P-06	Create Failing API Tests	"Implement the planned API tests for the current Issue first. Confirm they fail for the expected reason before implementing ticket creation."	Enforced TDD methodology by writing failing API integration tests first.
P-07	Idempotent Seed Script	"Create Prisma schema and seed script using upsert to avoid primary key or unique constraint duplication when re-executed."	Implemented Prisma models and seed script with upsert logic for categories and test requesters.
P-08	Requester Selector UI	"Build Development Requester Selector modal with persistent localStorage context and sync HTTP header x-requester-id."	Implemented React context provider, dropdown modal dialog, and request header interceptor.
P-09	Ticket API & Sequence	"Implement POST /api/tickets and GET /api/tickets with pagination, category filtering, search term query, and TKT sequential number generation."	Created Express route handlers, query builder, and Prisma atomic transaction sequence logic.
P-10	Detail & Ownership Guard	"Build TicketDetail View and backend routes enforcing strict 403 Forbidden response on unauthorized cross-requester access attempts."	Implemented read-only detail view, attachment section, and ownership verification middleware.

## Answer Part 5: Development Requester Selection Screen



Figure 5.1: Development Requester Selector Modal

- **Modal Behavior:** Prompts selection of an active testing Requester identity upon landing.
- **Context Persistence:** Selected Requester ID is saved in local state/context and sent via `x-requester-id` HTTP header for API requests.

## Answer Part 6: Working Ticket Screen: Create Mode (10 คะแนน)



Figure 6.1: Create Ticket Screen with Logged-in Requester populated



Figure 6.2: Field Validation Errors on Submitting Empty Form

- **Form Validation:** Validates required fields (Summary min 5 chars, Description min 10 chars, Category, Priority).
- **Reference Data:** Loads Category and Related System dropdown lists dynamically from backend.

## Answer Part 7: Working My Tickets Screen (10 คะแนน)



Figure 7.1: My Tickets Paginated List for Requester A



Figure 7.2: Text Search Filtering across Ticket Number and Summary

- **Data Isolation:** Only tickets belonging strictly to the currently selected Requester ID are returned.
- **Pagination:** Supports limit=10 items per page with page navigation buttons and total items counter.

## Answer Part 8: Ticket Screen View Mode & Attachments (5 คะแนน)



Figure 8.1: Ticket Detail Read-Only View with Attachment Section

Ownership Security Verification (403 Forbidden):

```
// Backend Ownership Guard Middleware (server/src/app.ts)
if (ticket.requesterId !== requesterId) {
 res.status(403).json({ error: "Forbidden: You are not the owner of this ticket" });
 return;
}
```

Answer Part 9: Zen Green UI and Responsive Evidence (5 คะแนน)

ลิงก์: <https://github.com/Suprawi5227/tokticket/blob/main/docs/lab-02/ui-spec.md>

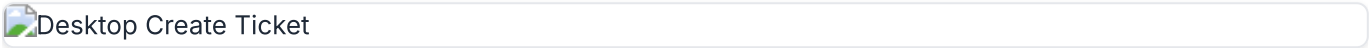
1. Design System Tokens & Color Palette

Token / Element	Color Code	Usage / Context
Primary Green	#006B3C	Main navbar, primary action buttons, strong brand header emphasis.
Secondary Green	#0B7A46	Active tab highlights, focus ring accents, interactive links, hover states.
Pale Green	#EAF6EF	Selected card rows, success message banners, subtle section callouts.
Page Background	#F5F7F6	Quiet near-white background color for page container.
Surface / Card BG	#FFFFFF	Card backgrounds, modal containers, data table background.
Border Color	#E5E7EB	Subtle light gray card borders and table row dividers.
Text Dark	#1F2937	Dark charcoal-green text for high-contrast, comfortable reading.
Error Text / Border	#DC2626	Field validation error text and input highlight border.

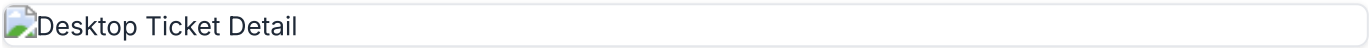
2. Desktop Viewport Screenshots (1280px)



Desktop My Tickets Table View

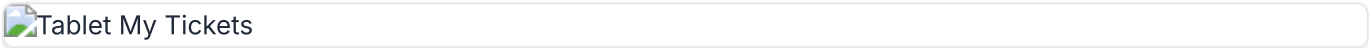


Desktop Create Ticket Form View



Desktop Ticket Detail View

3. Tablet Viewport Screenshots (768px)



*Tablet My Tickets View*

 Tablet Create Ticket

*Tablet Create Ticket View*

 Tablet Ticket Detail

*Tablet Ticket Detail View*

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#### **4. Mobile Viewport Screenshots (375px)**

 Mobile My Tickets

*Mobile My Tickets View*

 Mobile Create Ticket

*Mobile Create Ticket View*

 Mobile Ticket Detail

*Mobile Ticket Detail View*